Simulations of current sheet thinning and reconnection

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[1] Thin current sheet structures with a thickness comparable to the ion gyro-radius have been inferred from spacecraft observations during substorms. This study presents two-dimensional hybrid and Hall-MHD simulations of current sheet thinning and reconnection that include electron pressure tensor effects. Results demonstrate that reconnection occurs due to the electron pressure tensor effects as the sheet is compressed by the application of an out-of-plane electric field on the lobe boundaries. Results from hybrid and Hall-MHD calculations for the current sheet thinning process are compared. During the thinning period, the out-of-plane ion velocity from the hybrid simulation shows an enhancement at the edge of the sheet and a reduction at the sheet center. As reconnection proceeds, it becomes peaked along the Earth-tail direction away from the X point. In contrast, the out-of-plane ion velocity from the Hall-MHD simulation remains peaked at the center of the simulation. The ion off-diagonal pressure tensor effects are considered as the cause of the differing particle-ion and fluid-ion dynamics. The dynamics of current sheet thinning and reconnection is also modeled using a new integrated approach in which Hall-MHD calculations are embedded inside an MHD simulation. It is shown that the embedded simulation can provide smooth transitions of plasma and field quantities between the two regions, and the small-scale physics in the compressed current sheet and in the near-X-point region are well represented. *INDEX TERMS:* 7835 Space Plasma Physics: Magnetic reconnection; 7843 Space Plasma Physics: Numerical simulation studies; 2753 Magnetospheric Physics: Numerical modeling; 2764 Magnetospheric Physics: Plasma sheet; *KEYWORDS:* Magnetic reconnection, current sheet, numerical simulation studies

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1. Introduction

[2] Thin current sheet structures have been observed in the near-Earth, mid-tail, and distant tail regions during substorms [*Sergeev et al.*, 1993; *Pulkkinen et al.*, 1993, 1994, 1999; *Sanny et al.*, 1994]. It was inferred from the measurements that current sheets can thin to a thickness comparable to the ion gyro-radius (based on the magnetic field strength outside the sheet). These thin current sheet observations indicate a two-scale structure comprising a thin current sheet with enhanced current density embedded within a much thicker plasma sheet. Thin current sheets have also been observed recently at high latitudes, where the Polar spacecraft is at times able to remain in the sheet for extended periods of time [*Scudder et al.*, 2002].

[3] Examining the mechanisms involved in the current sheet thinning process is a subject of interest because of its importance to the understanding of the substorm dynamics. Various simulation studies, including full particle (kinetic ions and electrons), hybrid (kinetic ions and massless fluid electrons), Hall-magnetohydrodynamics (Hall-MHD), and large scale MHD simulations, have demonstrated that thin current sheets form as a result of the duskward external driving electric field imposed at the lobe boundaries [*Pritchett and Coroniti*, 1994, 1995; *Hesse et al.*, 1996; *Birn et al.*, 1998; *Rastätter et al.*, 1999]. During the compression of the sheet, the magnetic field component normal to the sheet reduces substantially as the magnetic flux is transfered to the tail. As a result, the thin current sheet configuration can become unstable and reconnection may be initiated [*Pritchett and Coroniti*, 1995].

[4] The aforementioned scale length of the thin current sheet and the two-scale structure inferred from observations have been clearly demonstrated in full particle simulations of thin current sheet formation [*Pritchett and Coroniti*, 1994, 1995]. It is difficult to perform a detailed study of the current carriers from spacecraft measurements of thin current sheets because of the coarse temporal resolution of the data [*Pulkkinen et al.*, 1994]. However, the simulation results of *Pritchett and Coroniti* [1994, 1995] have revealed that the enhanced current is carried by the electrons, which undergo $E \times B$ drifts in the electrostatic electric field normal to the sheet established during the course of thinning. These results were also confirmed later by hybrid and Hall-MHD simulations [*Hesse et al.*, 1996; *Rastätter et al.*, 1999]. The

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analysis of recent Polar spacecraft observations at high latitudes also demonstrates that electrons carry most of the current [*Scudder et al.*, 2002].

[5] In addition to studies of current sheet formation, extensive studies of the dynamics of thin current sheets, especially collisionless magnetic reconnection, have been conducted. Results of recent 2-D simulation studies, using various simulation methods and a Harris equilibrium, have demonstrated clearly the role of electron physics. From the electron momentum equation, the generalized Ohm's law is written as

$$\mathbf{E} = -\mathbf{v_i} \times \mathbf{B} + \frac{1}{en_e}(\mathbf{J} \times \mathbf{B} - \nabla \cdot \underline{\mathbf{P}}_e) - \frac{m_e d\mathbf{v_e}}{e\ dt} + \eta \mathbf{J}, \quad (1)$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, $\mathbf{J}$ the total current density, $\underline{\mathbf{P}}_e$ the electron pressure tensor, $\mathbf{v_i}$ the ion fluid velocity, $\mathbf{v_e}$ the electron fluid velocity, η the resistivity, n_e the electron density, m_e the electron mass, and e the electron charge. In order for reconnection to occur, the dissipative electric field near the X point is provided by the divergence of the electron pressure tensor at the ion inertial length scale c/ω_{pi} (the third term) [*Hesse and Winske*, 1994; *Kuznetsova et al.*, 1998, 2000], or by the electron inertia at the electron inertial scale c/ω_{pe} (the fourth term) [*Biskamp et al.*, 1997; *Drake et al.*, 1997; *Shay et al.*, 1999; *Rogers et al.*, 2000], where ω_{pi} (ω_{pe}) is the ion (electron) plasma frequency based on the density in the sheet. It has been found that the reconnection electric field from the electron pressure tensor dominates that of the electron inertia in thin current sheets of ion inertial scale [*Hesse and Winske*, 1998; *Kuznetsova et al.*, 1998, 2000; *Hesse et al.*, 1999; *Pritchett*, 2001]. The off-diagonal electron pressure tensor contributions have also been inferred from the data in the diffusion region [*Scudder et al.*, 2002]. Moreover, 2-D reconnection studies in a thin current sheet [*Birn et al.*, 2001] have demonstrated that inclusion of the Hall physics in the generalized Ohm's law (the second term in equation (1)) yields a fast reconnection rate, and that reconnection modeled by conventional resistive MHD simulations (including only the first and the last terms) produces a dramatically slower rate, unless a large localized or current-dependent resistivity is employed.

[6] Thus, fast collisionless reconnection can be studied using reduced models that exclude the effects of electron inertia but include the calculation of the full electron pressure tensor

$$\mathbf{E} = -\mathbf{v_i} \times \mathbf{B} + \frac{1}{en_e}(\mathbf{J} \times \mathbf{B} - \nabla \cdot \underline{\mathbf{P}}_e). \quad (2)$$

The advantage of this model is that it omits the electron skin depth scale (c/ω_{pe}) structure of the dissipation region. It is known that the c/ω_{pe} scale structure does not control the rate of reconnection [*Shay et al.*, 1998, 1999]; however, it demands high spatial resolution and therefore a much greater computing effort. Such reduced models include hybrid (particle ions) and Hall-MHD (two-fluid) calculations [*Hesse and Winske*, 1994; *Winske and Hesse*, 1994; *Yin et al.*, 2001a] that neglect the electron inertia but employ a reduced kinetic electron description. The essential electron physics near the X point can be represented well by resolving only ion spatial and temporal scales.

[7] In this study these reduced models are used to simulate the formation of thin current sheets in two spatial dimensions. Both 1-D and 2-D current sheet equilibria with a thicker current sheet are used and out-of-plane electric fields are applied at the lobe boundaries to compress the sheet. The boundary driving resembles the conditions that the magnetotail current sheet experiences during the growth phase of a substorm. We compare the ion motions from the hybrid and Hall-MHD calculations and identify what is missing with the two-fluid approach. We demonstrate that current sheet thinning from external driving can lead to a natural initiation of reconnection as the electron pressure tensor effects become important. Hall-MHD simulations with electron pressure tensor effects have not been used previously to address the current sheet thinning issues. Hybrid simulation studies of sheet thinning using the electron pressure tensor model have been conducted earlier. However, as noted by *Hesse et al.* [1996] these studies concern the dynamics of a large section of the magnetotail and the numerical dissipation associated with the coarse spatial resolution prevented the onset of physical reconnection to be observed in the simulations.

[8] Traditionally, plasma processes with very different spatial and temporal scales are studied separately. Large scale phenomena are typically modeled using MHD simulations, whereas small-scale processes near the X point require at least Hall-MHD calculations to capture the important physics. The dynamics of lobe regions can be well approximated by MHD calculations: Far from the diffusion region the effects of Hall current and electron pressure are insignificant, the ions and electrons follow the motion of the field lines, and $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$ (where $\mathbf{v_i} = \mathbf{v_e} = \mathbf{v}$) provides a good approximation. However, as the current sheet thins to a thickness on the order of the ion gyro-radius, the electron and ion motions become decoupled and the thin sheet region should be modeled by two-fluid calculations. While MHD and Hall-MHD simulations are both well established tools for studying these the large and small scale phenomena, attempts have not been made previously to develop a simulation tool for collisionless reconnection in which different sets of equations are solved in different regions (i.e., an "embedded simulation"). In this study we examine the possibility of the embedded simulation approach and present some initial results of our embedded Hall-MHD/MHD simulations for the current sheet thinning and reconnection problem.

2. Numerical Approach

2.1. Hybrid and Hall-MHD Simulation Codes

[9] Both the 2-D hybrid (particle ions, massless fluid electrons) and Hall-MHD (massless electrons) codes used in this study retain the effects of the electron pressure tensor. The time evolution of the full electron pressure tensor $\underline{\mathbf{P}}_e$ can be obtained from the second moment description of the Vlasov equation for the electrons. By neglecting effects of the heat flux, the time variation of the full electron pressure

tensor $\underline{\mathbf{P}}_e$ is computed using the following equation [*Hesse and Winske*, 1994]:

$$\frac{\partial \underline{\mathbf{P}}_e}{\partial t} = -\mathbf{v_e}\cdot\nabla\underline{\mathbf{P}}_e - \underline{\mathbf{P}}_e\nabla\cdot\mathbf{v_e} - \underline{\mathbf{P}}_e\cdot\nabla\mathbf{v_e} - (\underline{\mathbf{P}}_e\cdot\nabla\mathbf{v_e})^T - \frac{\Omega_e}{\tau}(\underline{\mathbf{P}}_e - p\underline{\mathbf{1}}). \quad (3)$$

In this expression, Ω_e is the electron gyrofrequency, the superscript T is the transpose of the matrix, and $p = \frac{1}{3}\mathrm{Tr}(\mathbf{P}_e)$. The last term on the right-hand side of this equation represents the reduction of the nongyrotropic (off-diagonal) elements of the electron pressure tensor and the isotropization of the three diagonal elements, and τ is a parameter corresponding to a gyrotropization/isotropization time of the order of electron cyclotron time.

[10] Equation (2) indicates that in a small, localized region near the X point, where the local magnetic field vanishes, spatial gradients of the off-diagonal electron pressure tensor elements $-(\partial P^e_{xz}/\partial x + \partial P^e_{yz}/\partial y)$ provide the reconnection electric field E_z (the scale size of this localized region is indicated by Plate 4c of *Yin et al.* [2001a]). This collisionless electron viscous effect plays the role of the conventional localized resistivity at the X point. The viscous effects do not lead to significant localized electron heating, which would occur in a resistive model with scalar electron pressure and would need to be conducted away from the X point by an electron heat flux [*Coroniti and Eviatar*, 1977] that is neglected in equation (3).

[11] The simulations are done in the (x, y) plane, in which $\hat{x}$ is the direction normal to the sheet and $\hat{y}$ is along the sheet. The basic numerical methods employed in these codes are described in detail elsewhere [*Yin et al.*, 2001a]. In brief, for the purpose of comparing results from the two codes, the Hall-MHD code was developed with numerical methods similar to those used in conventional hybrid codes [*Winske and Quest*, 1988; *Winske and Omidi*, 1993; *Winske and Hesse*, 1994; *Hesse and Winske*, 1994]. Spatial grid quantities in both codes are defined on the same set of staggered finite-difference meshes, where the magnetic field components are defined at the centers of the cell and all other quantities are defined at the cells' vertices. The time differencing follows a conventional staggered leapfrog method. In the hybrid simulation, **E** and **B** are advanced in time with a predictor-corrector scheme [*Winske and Quest*, 1986]. In the Hall-MHD code the magnetic field **B** is advanced in time using a fourth-order Runge-Kutta scheme with subcycling [*Winske and Quest*, 1988]. In each time step used to advance the fluid moments (velocity, density, and pressure), five to ten substeps are used, typically, to update the fields. In the Hall-MHD code a smoothing routine is used instead of the conventional second-order viscosity (i.e., $\sim\nu\nabla^2\mathbf{v}$), as done in the hybrid code. In our previous study of reconnection in a thin Harris sheet [*Yin et al.*, 2001a] a sixth order hyperviscous dissipation, $\sim\nu(\nabla_x^6 + \nabla_y^6)\mathbf{B}$, was employed in the magnetic field equation by operator splitting and fast Fourier transform (fft) techniques to damp out fluctuations on very short spatial scales. In that study, the initial current sheet was thin (with a half thickness of 0.4 ion inertial length based on the asymptotic plasma density at the lobe) and a perturbation was imposed to the Harris equilibrium to initiate reconnection.

[12] Here, for the purpose of simulating current sheet thinning and reconnection, a thicker current sheet (half thickness of two or four times the ion inertial length) is used for the initial state and a driving electric field is applied at the lobe boundaries to compress the sheet. As the sheet thins, reconnection proceeds as a consequence of the electron pressure tensor. Because of the lobe boundary driving, some modifications are made in both codes to satisfy the new magnetic field boundary conditions. In particular, a fourth order hyperviscous dissipation, $\sim\nu(\nabla_x^4 + \nabla_y^4)\mathbf{B}$, is used in the magnetic field equation to keep the numerics tractable. It is implemented by band-diagonal (in the x dimension) and sine/cosine transform (in the y dimension) methods. This hyperviscous dissipation technique is also adopted in our hybrid code. It is known that of the various hybrid simulation methods the predictor-corrector scheme (employed by our hybrid code) is the most widely used because it gives very good energy conservation and it is rather robust, but there can be significant amount of whistler noise [*Winske et al.*, 2002]. These whistler fluctuations are on very short spatial scales and need to be damped out, especially for the boundary driving simulations, in order to stabilize the code. We have, for the first time, implemented the aforementioned hyperviscous dissipation in a hybrid code and found that it works effectively.

2.2. Method of Embedding

[13] As a first attempt to investigate the micro- macro-scale coupling in the magnetotail dynamics, we developed a new integrated approach in which Hall-MHD calculations with full electron pressure tensor model are embedded inside an MHD simulation near the X point. A schematic illustration of the approach is included in Figure 1. The Hall-MHD calculations run on the inner region to model correctly the ion inertial scale physics near the X point, including the decoupling of the ion and electron motions and the reconnection electric field produced by the off-diagonal electron pressure tensor terms. The MHD code runs on the outer region and calculates large scale fields and one-fluid quantities. The MHD solutions also provide appropriate boundary conditions for the two-fluid calculations done in inner region.

[14] In addition to the electric field and electron pressure tensor equations (2) and (3), the Hall-MHD code solves the continuity equation

$$\frac{\partial n}{\partial t} = -\nabla(n\mathbf{v_i}), \quad (4)$$

the ion momentum equation

$$nm_i\frac{\partial \mathbf{v_i}}{\partial t} = -nm_i(\mathbf{v_i}\cdot\nabla)\mathbf{v_i} - \nabla P_i - \nabla\cdot\underline{\mathbf{P}}_e + \frac{\mathbf{J}\times\mathbf{B}}{c}, \quad (5)$$

and the scalar ion pressure equation

$$\frac{\partial P_i}{\partial t} = -\nabla\cdot(\mathbf{v_i}P_i) - (\gamma - 1)P_i(\nabla\cdot\mathbf{v_i}), \quad (6)$$

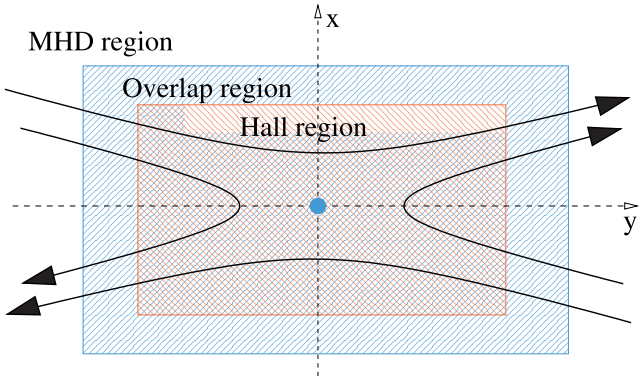


Figure 1. An illustration of the embedded approach for modeling magnetic reconnection in which Hall-MHD calculations are embedded inside an MHD simulation in the region near the X point.

where $\gamma = 5/3$ is the ratio of the specific heats. The Ampere's law in low frequency limit without the displacement current is

$$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{J}, \tag{7}$$

and Faraday's law is

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}. \tag{8}$$

In the MHD code, the electric field is computed simply from $\mathbf{E} = -\mathbf{v_i} \times \mathbf{B}$, and the electron pressure effects are not computed.

[15] An overlap region is included so that the transition between the two types of calculations is smooth. In this region the electric field is calculated from

$$\mathbf{E} = -\mathbf{v_i} \times \mathbf{B} + a(x,y)\mathbf{E}' \tag{9}$$

where

$$\mathbf{E}' = (1/en_e)(\mathbf{J} \times \mathbf{B} - \nabla \cdot \underline{\mathbf{P}}_e), \tag{10}$$

and

$$a(x,y) = (1/16)\{1 + \mathrm{erf}[\lambda_x(x - x_1)]\}\{1 + \mathrm{erf}[\lambda_x(x_2 - x)]\}$$
$$\{1 + \mathrm{erf}[\lambda_y(y - y_1)]\}\{1 + \mathrm{erf}[\lambda_y(y_2 - y)]\} \tag{11}$$

where x_1 and x_2 are the lower and upper boundaries of the Hall-MHD region and y_1 and y_2 are the left and right boundaries. The expression with the error function erf [*Abramowitz and Stegun*, 1970] ensures that $a(x, y) = 1$ in the Hall-MHD region and $a(x, y) = 0$ in the MHD region. The width of the transition region is determined by constants λ_x and λ_y in x and y directions, respectively. However, the introduction of the function $a(x, y)$ results in a magnetic field $\mathbf{B} \sim \nabla a(x, y) \times \mathbf{E}'$, including a component normal to the sheet, that requires additional care, such as smoothing, to remove.

3. Current Sheet Thinning and Reconnection

3.1. Initial Conditions

[16] The calculations employ an initial 2-D equilibrium [*Birn et al.*, 1975; *Lembège and Pellat*, 1982; *Birn et al.*, 1998]:

$$B_x = B_o F'(y)\{-(x - l_x/2)\tanh\{F(y)[x - l_x/2]/\alpha\} + \alpha/F(y)\} \tag{12}$$

$$B_y = B_o F(y)\tanh\{F(y)[x - l_x/2]/\alpha\}, \tag{13}$$

where B_o is the lobe field strength, α is the half-thickness of the current sheet, l_x is the width of the simulation domain, and

$$F(y) = (1 + \varepsilon y/\alpha)^{-\nu}. \tag{14}$$

In the case of a Harris sheet [*Harris*, 1962], $F(y) = 1$ and $B_x = 0$ (section 3.2). Since observations indicate that thin current sheets form during substorms in midtail and distant tail regions as well as in the near-Earth region [*Sergeev et al.*, 1993; *Pulkkinen et al.*, 1993, 1994, 1999; *Sanny et al.*, 1994], dipole fields are not included in our magnetic field configuration.

[17] The hybrid calculation includes two ion components: sheet ions and uniform background ions. Both ion populations have the same temperature so that the hybrid and Hall-MHD simulations can be performed under the same initial conditions. The hybrid run uses 20 background ions per grid and an average of 20 sheet ions per grid. The lobe ion plasma β is 0.2 and $T_i/T_e = 5$, so the ions are assumed to be the main current carriers initially.

[18] An external electric field E_z normal to the simulation plane is imposed on the x boundaries to simulate the thinning process during the growth phase of a substorm. After an initial time interval when the sheet is compressed substantially, the driver is removed. The imposed E_z on the x boundaries varies as a function of y. Its maximum strength corresponds to an inflow speed of 0.1 v_A (v_A is the lobe Alfvén speed). Under the driving field, the lobe magnetic field strength increases to compress the sheet.

3.2. Comparison of Hybrid and Hall-MHD Simulations

[19] To compare the results from hybrid and Hall-MHD simulations, we initially use a Harris equilibrium. The initial current sheet half thickness is $\alpha = 2c/\omega_{pi}$, where for all the simulations the length scale c/ω_{pi} is based on the asymptotic plasma density at the lobe; this half thickness is equivalent to 6.3 ion gyro-radii in the lobe field. The spatial domain of 10×10 $(c/\omega_{pi})^2$ consists of 128×128 grid points. An external electric field $E_z(y) \sim \cos[\pi(y - 5)/10]$ is imposed on the x boundaries.

[20] Results from the two types of simulations regarding the electron physics are in general agreement. A thin current sheet forms due to the external compression, as found in ideal MHD simulations [*Birn et al.*, 1998]. The thin current sheet thickness is on the order of an ion gyro-radius, and it is embedded within a thicker plasma sheet. The thin current

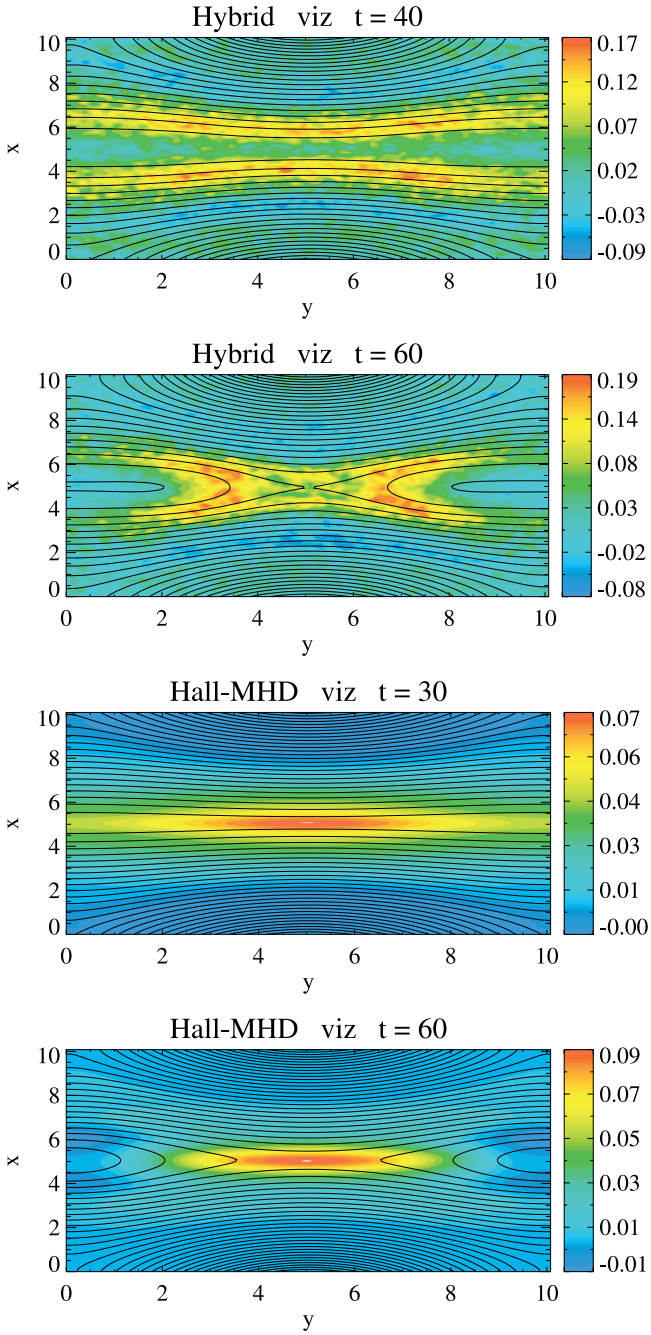


Figure 2. Comparison of the out-of-plane ion velocity v_z from hybrid and Hall-MHD simulations. v_z is plotted on a linear scale in units of v_A; overlaid are the magnetic field lines in the simulation plane.

sheet current is predominantly carried by electrons. These results have been discussed in previous studies using full particle, hybrid, and Hall-MHD simulations [*Pritchett and Coroniti*, 1994, 1995; *Hesse et al.*, 1996; *Rastätter et al.*, 1999].

[21] Here we present differences between the hybrid and Hall-MHD simulations resulting from the dynamics of the out-of-plane ion velocity v_z. Figure 2 shows a comparison of the ion velocity v_z from hybrid and Hall-MHD simulations. Both configurations during the driving (hybrid at $t\ \Omega_{ci} = 40$ and Hall-MHD at $t\ \Omega_{ci} = 30$) and during subsequent reconnection (hybrid and Hall-MHD both at $t\ \Omega_{ci} = 60$) are displayed. During the driving period, the global structure of v_z from the hybrid simulation (top panel) shows an enhancement of v_z at the edge of the sheet and a reduction at the center of the sheet. As reconnection proceeds, v_z from the hybrid calculation (second panel from top) becomes peaked along the y direction away from the X point. In contrast, the spatial configuration of v_z from the Hall-MHD run remains peaked at the center of the simulation during both the driving period (third panel from top) and recon-

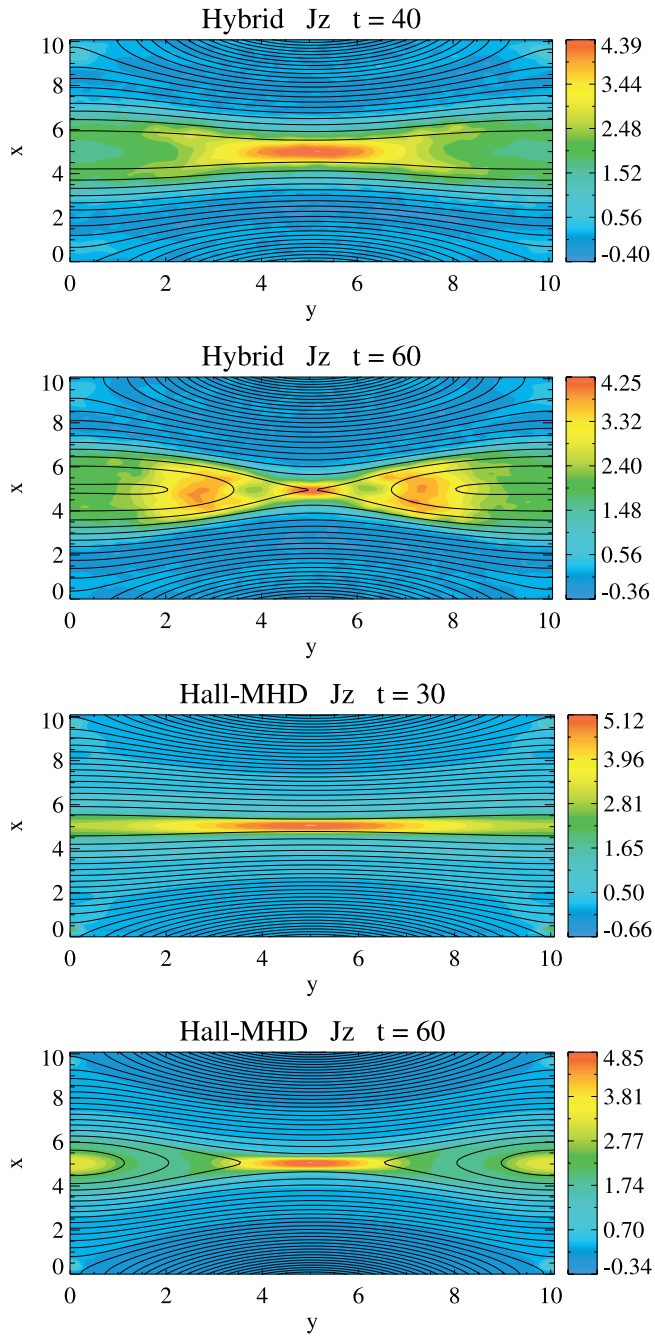
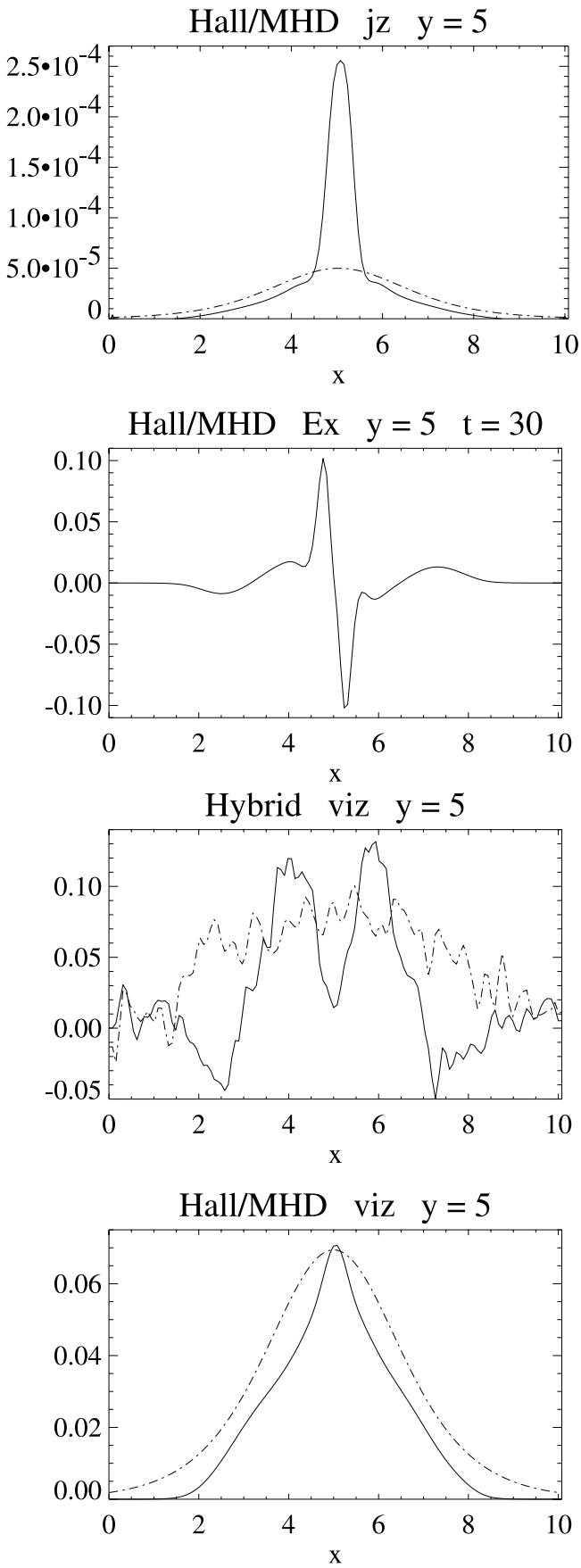


Figure 3. The out-of-plane current density (normalized to the maximum value at $t\ \Omega_{ci} = 0$) from hybrid and Hall-MHD simulations.



nection (bottom panel). The difference in the v_z transport for the particle ions and fluid ions is similar to that found in the studies of 2-D reconnection in thin current sheets [*Yin et al.*, 2001b, 2002].

[22] Although the two codes retain the same electron physics, the difference in the ion v_z leads to a difference in the configuration of the out-of-plane current density J_z, as presented in Figure 3. Since the ion v_z become peaked at the edge of the sheet, its effect is to broaden the cross-sheet current profile while the external driving compresses the sheet. As a consequence, a longer driving time is applied to the hybrid run for the onset of reconnection to occur.

[23] Cross-sheet profiles from the hybrid and Hall-MHD simulations are also examined. The upper two panels in Figure 4 show the current density J_z at $y = 5$ (solid curve for $t\,\Omega_{ci} = 30$; broken curve for $t\,\Omega_{ci} = 0$) and the electric field E_x at $y = 5$ ($t\,\Omega_{ci} = 30$) from the Hall-MHD run. The thinning process produces an inward directing electric field E_x normal to the sheet. The electron $E_x \times B_y$ drift as well as the electron diamagnetic drift under the pressure force result in an enhanced thin current sheet current density that is carried predominantly by the electrons, as pointed out earlier by *Pritchett and Coroniti* [1994, 1995]. The cross-sheet scale length of the thin current is shown to be consistent with that of E_x field. Displayed in the lower two panels are the ion v_z profile at $y = 5$ from hybrid ($t\,\Omega_{ci} = 40$, solid curve) and Hall-MHD simulation ($t\,\Omega_{ci} = 30$, solid curve); the broken curves indicate initial ion v_z for comparison. The hybrid simulation shows that the ion v_z at the center of the sheet decreases as the sheet is compressed and v_z becomes peaked along the edge of the sheet. If the ions also drift in the E_x field (assuming ions are magnetized) the effect of the ion $E_x \times B_y$ drift should result in a reduction of the initial ion diamagnetic drift. Both the $E \times B$ drift and the diamagnetic drift under pressure force are fluid quantities and are retained in both the hybrid and Hall-MHD calculations. However, the cross-sheet profile of v_z from the Hall-MHD simulation does not evolve in time the same way as that from the hybrid run.

[24] We expect the ion behavior seen in the hybrid simulation to be due to ion gyro-radius effects. As a test, ions are run in the Hall-MHD fields. Initially, the test ions are in pressure equilibrium with the magnetic field. As the Hall-MHD run proceeds, the test ions are advanced in the electric and magnetic fields but they are not fed back into the fluid calculation. The ion dynamics from the Hall-MHD/test ion run are shown in Figure 5. The out-of-plane ion velocity v_z during thinning and during reconnection displayed in the upper two panels resemble that from the

Figure 4. (opposite) Cross-sheet profiles from hybrid and Hall-MHD simulations. From top to bottom: Current density J_z from Hall-MHD run ($t\,\Omega_{ci} = 30$, solid curve), electric field E_x at $y = 5$ from Hall-MHD run ($t\,\Omega_{ci} = 30$), ion velocity v_z at $y = 5$ from hybrid simulation ($t\,\Omega_{ci} = 40$, solid curve), and ion velocity v_z at $y = 5$ from Hall-MHD run ($t\,\Omega_{ci} = 30$, solid curve). The initial profiles are also shown by the broken curves in the top, third, and bottom panels for comparison. J_z is in arbitrary units; E_x is normalized to $v_A B_o/c$ where v_A is the lobe Alfvén speed, and velocity is in units of v_A.

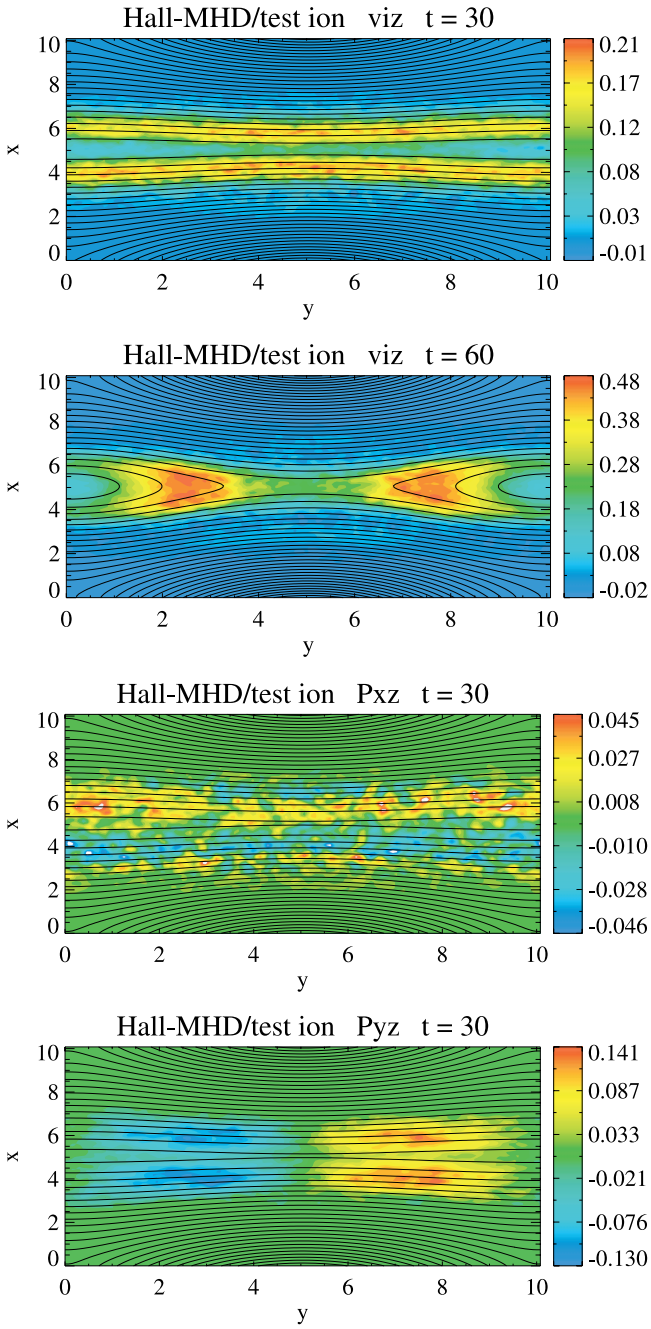


Figure 5. The out-of-plane ion velocity v_z and off-diagonal ion pressure tensor terms P^i_{xz} and $P_{yz}{}^i$ from a Hall-MHD/test ion run. Velocity is in units of v_A; the pressure tensor terms are averaged over an ion gyro-period and are normalized to the initial lobe ion pressure.

hybrid simulation shown in Figure 2. We computed the off-diagonal ion pressure tensor terms P^i_{xz} and P^i_{yz} from the test ions. These two terms (normalized to the initial lobe ion pressure and averaged over an ion gyro-period) are shown in the lower two panels of Figure 5. Both terms show a characteristic global structure: P^i_{xz} peaks along the edge of the sheet whereas the maximum strength of P^i_{yz} is localized in regions of the sheet away from the driving maximum (at $y = 5$). The P^i_{xz} term has a smaller magnitude than P^i_{yz}, and its spatial structure is noisy due to the much smaller number of ions at the edge of the sheet.

[25] The ion gyro-radius effects are contained in these off-diagonal ion pressure tensor terms. If one replaces the scalar pressure in equation (5) with the pressure tensor term $-\nabla \cdot \underline{\mathbf{P}}_i$, it results in a component in the z direction which is $-(\partial P^i_{yz}/\partial y + \partial P^i_{xz}/\partial x)$. Thus, effects of the ion gyro-orbit can modify the ion momentum transport in the z direction. A method that uses test ion distributions to evaluate these off-diagonal ion pressure tensor terms and to implement the ion gyro-radius corrections in Hall-MHD simulations has

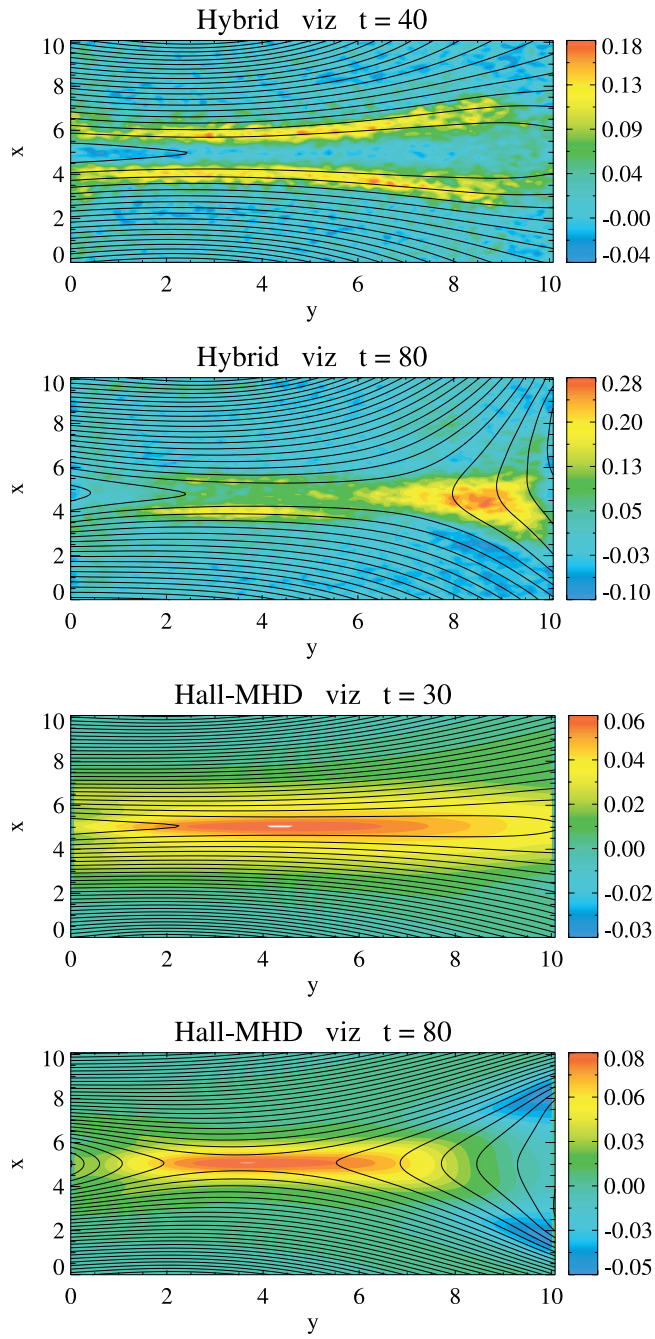


Figure 6. Comparison of the out-of-plane ion velocity v_z from hybrid and Hall-MHD simulations using a 2-D equilibrium.

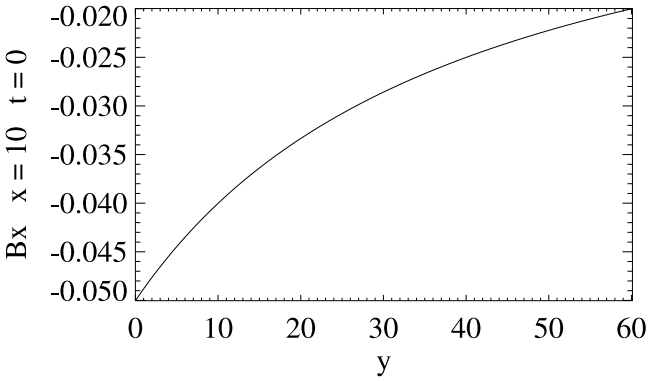


Figure 7. The initial normal magnetic field profile along y at the center of the sheet (x = 10). B_x is normalized to the lobe field strength.

been examined for 2-D reconnection in thin current sheets [*Yin et al.*, 2001b, 2002]. It was shown that the modified Hall-MHD calculations with the ion pressure tensor effects (mainly from the P^i_{yz} term) allow the out-of-plane ion momentum to move away from the X point during reconnection, similar to the hybrid result shown by the second panel in Figure 2. We speculate that the configuration of ion v_z during the compression of the sheet seen in the top panel of Figure 2 is produced because of the P^i_{xz} term. However, with the much smaller number of ions at the edge of the sheet, it is difficult to implement the P^i_{xz} term into the ion momentum equation using test particles.

[26] We have also carried out hybrid and Hall-MHD simulations of reconnection initiated with a 2-D equilibrium, as shown in Figure 6. As in the previous calculation, the system size is relatively small (10×10 $(c/\omega_{pi})^2$). The values of $\varepsilon = 0.1$ and $\nu = 0.5$ (see equation (14)) are used for the initial equilibrium, and the resulting initial value of B_x/B_o in the center of the sheet at the left boundary is about 5% of the lobe field strength. The maximum driving of E_z is imposed on the x boundaries at $y = 2.5c/\omega_{pi}$, and its profile varies as $\cos[0.65\pi(y - 2.5)/10]$. The y-boundary conditions $db_x/dy = 0$ and $db_y/dy = 0$, where b_x and b_y are the perturbed magnetic fields, are used for these runs. We again compare the out-of-plane ion velocity v_z at several times in each calculation. Figure 6 shows that during the driving period ($t\ \Omega_{ci} = 40$ is displayed for the hybrid run and $t\ \Omega_{ci} = 30$ for the Hall-MHD run) v_z from the hybrid simulation (top panel) is enhanced at the edge of the sheet and reduced at the center of the sheet. In contrast, v_z from the Hall-MHD run (third panel from top) remains peaked at the center of the simulation. In a reconnected geometry ($t\ \Omega_{ci} = 80$), v_z from the hybrid calculation (second panel from top) becomes peaked to the right of the X point along the y direction, whereas the peak of v_z from the Hall-MHD run (bottom panel) is centered at the X point. The differences between hybrid and Hall-MHD simulations are similar to those seen earlier in Figure 2.

3.3. Embedding of Hall-MHD Calculations Within a MHD Simulation

[27] We have discussed in section 3.2 that, in comparison with the hybrid calculations, the Hall-MHD simulation with the scalar ion pressure cannot describe the kinetic effects in the ion out-of-plane motion and the correct ion current profile. Nevertheless, the Hall-MHD simulation with the full electron pressure model is capable of including the essential electron physics near the X point for the current sheet thinning and reconnection modeling. The external driving perturbations applied at the lobe boundaries are communicated to the inner region by fast compressional waves which are represented in both MHD and Hall-MHD calculations. Thus, one can embed Hall-MHD calculations inside an MHD simulation to simulate current sheet thinning and reconnection. In this section we present some initial results of the embedded simulation effort.

[28] The initial equilibrium is 2-D with a finite normal component B_x with $\varepsilon = 0.1$ and $\nu = 0.5$. The initial value of B_x/B_o in the center of the sheet at the left boundary is about 5% of the lobe field strength and its profile along y at the center of the sheet (x = 10) is shown in Figure 7. The spatial domain is 60×20 $(c/\omega_{pi})^2$, much larger than the calculations presented earlier in this paper, and consists of 512×256 grid points. The initial current sheet half thickness at the left boundary is $\alpha = 4c/\omega_{pi}$, which corresponds to about 12.6 ion gyro-radii in the lobe field. The profile of the imposed E_z on the x boundaries varies as $\cos[0.66\pi(y - 15)/60]$ (i.e., the maximum driving is at $y = 15c/\omega_{pi}$). A comparison of the in-plane magnetic field and out-of-plane current density J_z from the Hall-MHD and embedded Hall-MHD/MHD simulations is displayed in Figure 8 (the y-boundary conditions $b_x = 0$ and $db_y/dy = 0$, where b_x and b_y are the perturbed magnetic fields, are used).

[29] The initial 2-D equilibrium is shown in the top panel. The configurations from the two calculations during reconnection (at $t\ \Omega_{ci} = 190$) after the compression of the sheet under the boundary driving are shown in the middle and bottom panels. For the embedded simulation shown in the bottom panel, the inner Hall-MHD region extends from x = 5.5 to x = 14.5 and y = 0 to y = 50. The results indicate that the transition from the Hall-MHD to the MHD region is smooth. Both calculations show that as the sheet thins the current density increases and the normal magnetic field B_x decreases. At the present stage of the embedded code, the grid size in the MHD region is the same as that in the Hall-MHD region. It is straightforward, however, to modify the embedded code so that the MHD region uses larger grids to eliminate unnecessary calculations on the fine mash.

[30] Figure 9 shows the reconnected flux as a function of time from the Hall-MHD and embedded Hall-MHD/MHD simulations. The flux is computed from a time integration of the out-of-plane electric field, $\int E_z dt$, where E_z is averaged over nine grids near the X point. It is seen that the two curves are indistinguishable, which demonstrates the excellent agreement between the two simulation runs.

4. Summary

[31] We have presented 2-D simulations of current sheet thinning and reconnection using hybrid and Hall-MHD codes that include the electron pressure tensor effects. Unlike studies of reconnection in thin current sheets, in which reconnection is often initiated by an imposed initial perturbation, we have applied an out-of-plane electric field on the lobe boundaries to compress an initially thick current

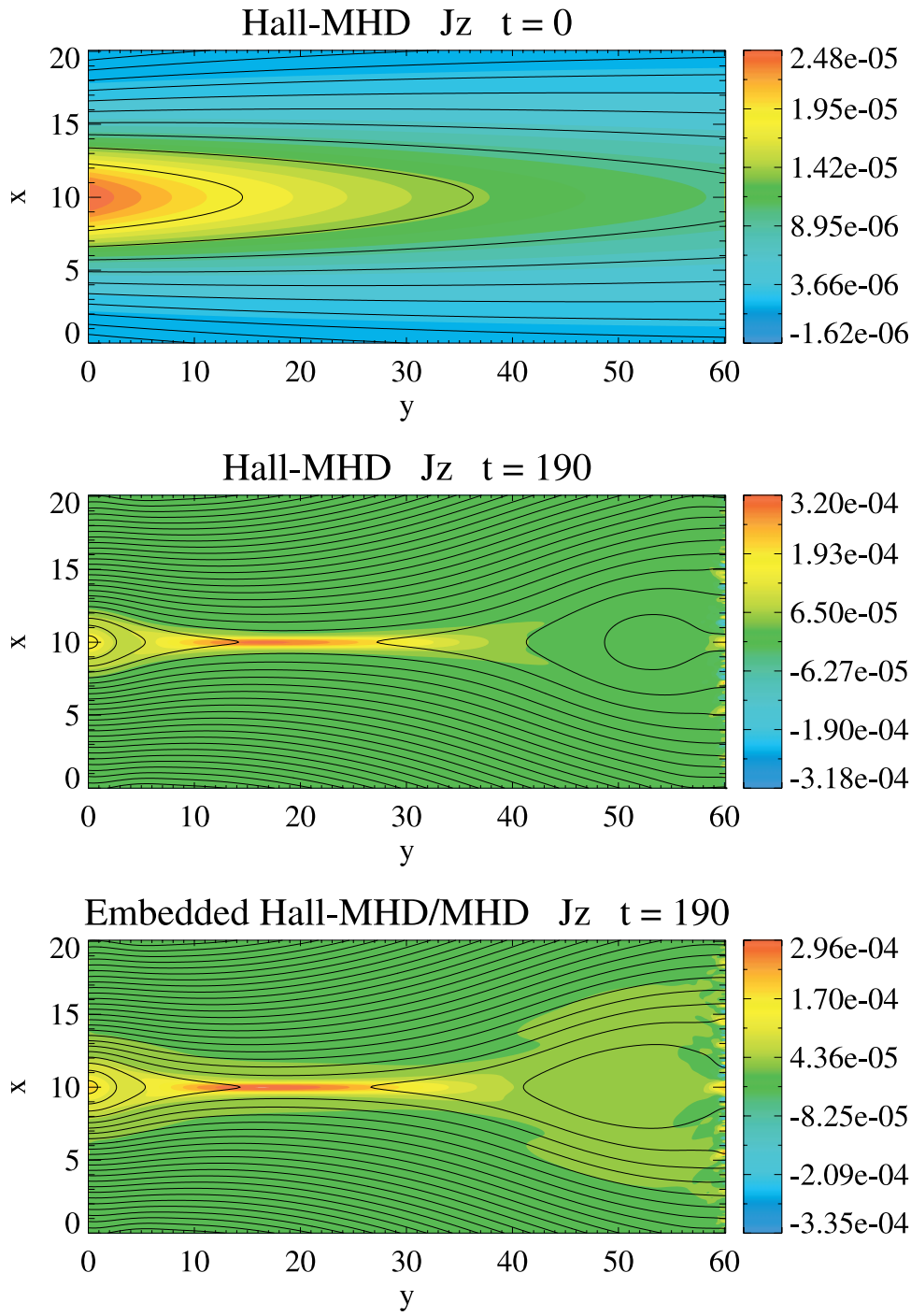


Figure 8. Comparison of Hall-MHD and embedded Hall-MHD/MHD simulations. Overlaid on the magnetic field lines in the simulation plane are color contours of J_z plotted on a linear scale (in arbitrary units). Spatial coordinates x and y are in units of c/ω_{pi}. For the embedded simulation shown in the bottom panel, the inner Hall-MHD region extends from $x = 5.5$ to $x = 14.5$ and $y = 0$ to $y = 50$.

sheet. Results have demonstrated that reconnection occurs due to the electron pressure tensor effects as the sheet thins to a thickness below the ion initial length. In both codes a fourth order hyperviscous dissipation is used in the magnetic field equation to damp out fluctuations on very short spatial scales. The implementation of this technique in the hybrid code is crucial for the stability of the boundary driving simulation with high spatial resolution. High spatial resolution is required in order to observe reconnection due to electron pressure tensor effects rather than numerical dissipation caused by a low spatial resolution.

[32] We have compared results from hybrid and Hall-MHD calculations for the current sheet thinning process. As reported in previous simulation studies [*Pritchett and Coroniti*, 1994, 1995; *Hesse et al.*, 1996; *Rastätter et al.*, 1999], for the compressed current sheet both codes yield a two-

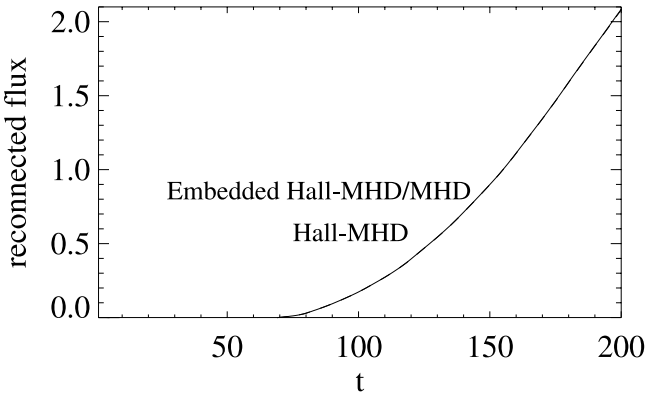


Figure 9. Reconnected flux as a function of time (time is normalized to $t\ \Omega_{ci}$) from Hall-MHD and embedded Hall-MHD/MHD simulations of current sheet thinning and reconnection. The curves are identical.

scale structure consisting of thin current sheet embedded within a thicker plasma sheet, and the dominant current carriers are the electrons. However, the cross-field ion transport during the driving period and during the subsequent reconnection appears to be different in the two calculations. During the driving period, the out-of-plane ion velocity v_z from the hybrid simulation shows an enhancement at the edge of the sheet and a reduction at the center of the sheet; as reconnection proceeds, v_z becomes peaked along the y direction away from the X point. In contrast, the v_z from the Hall-MHD simulation remains peaked at the center of the simulation during both the driving period and reconnection.

[33] The off-diagonal ion pressure tensor terms P^i_{xz} and P^i_{yz} were computed using test ions. We found that P^i_{xz} peaks along the edge of the sheet whereas the maximum strength of P^i_{yz} is localized in the central region of the sheet away from the driving maximum. Our previous simulations of reconnection in a thin current sheet [*Yin et al.*, 2001b, 2002] indicate that the ion pressure term P^i_{yz} allows for the cross-field transport of v_z from the X point during reconnection. We thus suggest that during the compression of the sheet the ion v_z motion seen in the hybrid calculations could be described in fluid calculations by including effects of the P^i_{xz} term in the ion momentum equation.

[34] Using the embedded simulation approach for the current sheet thinning and reconnection study, we have also carried out a first step towards the micro- macro-scale coupling in the modeling of the magnetotail dynamics. We have found that a Hall-MHD model embedded in an MHD calculation can provide smooth transitions of plasma and field quantities between the two regions. Furthermore, the small-scale physics in the compressed current sheet and in the near-X-point region are well represented by Hall-MHD, while the external MHD model provides appropriate, time-dependent boundary conditions. The embedded simulation approach has broad applicability to other systems characterized by multiple spatial and temporal scales and, therefore, is a topic deserving the investment of future research efforts.

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References

Abramowitz, M., and I. A. Stegun, *Handbook of Mathematical Functions*, Dover, Mineola, N.Y., 1970.

Birn, J., R. Sommer, and K. Schindler, Open and closed magnetospheric tail configurations and their stability, *Astrophys. Space Sci.*, *35*, 389, 1975.

Birn, J., M. Hesse, and K. Schindler, Formation of thin current sheets in space plasmas, *J. Geophys. Res.*, *103*, 6843, 1998.

Birn, J., et al., Environment Modeling (GEM) magnetic reconnection challenge, *J. Geophys. Res.*, *106*, 3715, 2001.

Biskamp, D., E. Schwarz, and J. F. Drake, Two-fluid theory of collisionless magnetic reconnection, *Phys. Plasmas*, *4*, 1002, 1997.

Coroniti, F. V., and A. Eviatar, Magnetic field reconnection in a collisionless plasma, *Ap. J. Suppl.*, *33*, 189, 1977.

Drake, J. F., D. Biskamp, and A. Zeiler, Breakup of the electron current layer during 3-D collisionless magnetic reconnection, *Geophys. Res. Lett.*, *24*, 2921, 1997.

Harris, E. G., On a plasma sheath separating regions of oppositely directed magnetic field, *Nuovo Cimento*, *23*, 115, 1962.

Hesse, M., and D. Winske, Hybrid simulations of collisionless reconnection in current sheets, *J. Geophys. Res.*, *99*, 11,177, 1994.

Hesse, M., and D. Winske, Electron dissipation in collisionless magnetic reconnection, *J. Geophys. Res.*, *103*, 26,479, 1998.

Hesse, M., D. Winske, M. Kuznetsova, J. Birn, and K. Schindler, Hybrid modeling of the formation of thin current sheets in magnetotail configurations, *J. Geomag. Geoelectr.*, *48*, 749, 1996.

Hesse, M., K. Schindler, J. Birn, and M. Kuznetsova, The diffusion region in collisionless magnetic reconnection, *Phys. Plasmas*, *6*, 1781, 1999.

Kuznetsova, M. M., M. Hesse, and D. Winske, Kinetic quasi-viscous and bulk flow inertia effects in collisionless magnetotail reconnection, *J. Geophys. Res.*, *103*, 199, 1998.

Kuznetsova, M. M., M. Hesse, and D. Winske, Toward a transport model of collisionless magnetic reconnection, *J. Geophys. Res.*, *105*, 7601, 2000.

Lembège, B., and R. Pellat, Stability of a thick two-dimensional quasineutral sheet, *Phys. Fluids*, *25*, 1995, 1982.

Pritchett, P. L., Geospace Environment Modeling magnetic reconnection challenge: Simulations with a full particle electromagnetic code, *J. Geophys. Res.*, *106*, 3783, 2001.

Pritchett, P. L., and F. V. Coroniti, Convection and the formation of thin current sheets in the near-Earth plasma sheet, *Geophys. Res. Lett.*, *21*, 1587, 1994.

Pritchett, P. L., and F. V. Coroniti, Formation of thin current sheets during plasma sheet convection, *J. Geophys. Res.*, *100*, 23,551, 1995.

Pulkkinen, T. I., D. N. Baker, C. J. Owen, J. T. Gosling, and N. Murthy, Thin current sheets in the deep magnetotail, *Geophys. Res. Lett.*, *20*, 2427, 1993.

Pulkkinen, T. I., D. N. Baker, D. G. Mitchell, R. L. McPherron, C. Y. Huang, and L. A. Frank, Thin current sheets in the magnetotail during substorms: CDAW 6 revisited, *J. Geophys. Res.*, *99*, 5793, 1994.

Pulkkinen, T. I., D. N. Baker, L. L. Cogger, L. A. Frank, J. B. Sigwarth, S. Kokubun, T. Mukai, H. J. Singer, J. A. Slavin, and L. Zelenyi, Spatial extent and dynamics of a thin current sheet during the substorm growth phase on December 10, 1996, *J. Geophys. Res.*, *104*, 28,475, 1999.

Rastätter, L., M. Hesse, and K. Schindler, Hall-MHD modeling of near-Earth magnetotail current sheet thinning and evolution, *J. Geophys. Res.*, *104*, 12,301, 1999.

Rogers, B. N., J. F. Drake, and M. A. Shay, The onset of turbulence in collisionless magnetic reconnection, *Geophys. Res. Lett.*, *27*, 3157, 2000.

Sanny, J., R. L. McPherron, C. T. Russell, D. N. Baker, T. I. Pulkkinen, and A. Nishida, Growth-phase thinning of the near-Earth current sheet during the CDAW 6 substorm, *J. Geophys. Res.*, *99*, 5805, 1994.

Scudder, J. D., F. S. Mozer, N. C. Maynard, and C. T. Russell, Fingerprints of collisionless reconnection at the separator, I, Ambipolar-Hall signatures, *J. Geophys. Res.*, *107*(A10), 1294, doi:10.1029/2001JA000126, 2002.

Sergeev, V. A., D. G. Mitchell, C. T. Russell, and D. J. Williams, Structure of the tail plasma/current sheet at $\sim 11 R_E$ and its changes in the course of a substorm, *J. Geophys. Res.*, *98*, 17,345, 1993.

Shay, M. A., J. F. Drake, R. E. Denton, and D. Biskamp, Structure of the dissipation region during collisionless magnetic reconnection, *J. Geophys. Res.*, *103*, 9165, 1998.

Shay, M. A., J. F. Drake, B. N. Rogers, and R. E. Denton, The scaling of collisionless, magnetic reconnection for large systems, *Geophys. Res. Lett.*, *26*, 2163, 1999.

Winske, D., and M. Hesse, Hybrid modeling of magnetic reconnection in space plasmas, *Physica D*, *77*, 268, 1994.

Winske, D., and N. Omidi, Hybrid codes: Methods and applications, in *Computer Space Plasma Physics: Simulation Techniques and Software*, p. 103, edited by H. Matsumoto and Y. Omura, Terra Sci., Tokyo, 1993.

Winske, D., and K. B. Quest, Electromagnetic ion beam instabilities: Comparison of one- and two-dimensional simulations, *J. Geophys. Res.*, *91*, 8789, 1986.

Winske, D., and K. B. Quest, Magnetic field and density fluctuations at perpendicular supercritical collisionless shocks, *J. Geophys. Res.*, *93*, 9681, 1988.

Winske, D., L. Yin, and N. Omidi, Hybrid codes: Past, present and future, in *Space Plasma Simulations*, in press, Springer-Verlag, New York, 2002.

Yin, L., D. Winske, S. P. Gary, and J. Birn, Hybrid and Hall-MHD simulations of collisionless reconnection: Dynamics of the electron pressure tensor, *J. Geophys. Res.*, *106*, 10,761, 2001a.

Yin, L., D. Winske, S. P. Gary, and J. Birn, Particle Hall-MHD simulation of collisionless reconnection: Ion gyro-radius correction, *Geophys. Res. Lett.*, *28*, 2173, 2001b.

Yin, L., D. Winske, S. P. Gary, and J. Birn, Hybrid and Hall-magnetohydrodynamics simulations of collisionless reconnection: Effect of the ion pressure tensor and particle Hall-magnetohydrodynamics, *Phys. Plasmas*, *9*, 2575, 2002.

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